ITA03-MOBILE COMPUTING ASSIGNMENT-1

ASSIGNMENT 1 – CO1

CO1 Coverage: Cellular network concepts; Frequency reuse and interference in cells; Access channels and mobile network capacity; Fundamentals of mobile computing

Nature of Assignment: Individual
Duration: 1 Week
Total Marks: 50
Submission Date: 04/11/26

Assignment Title: Design, Analyse and Optimise a Hexagonal Cellular Network for Spectrum Allocation, Interference Management and Scalable Coverage Planning in a University Township (Total Marks – 50)

Duration:1week

Scenario — Cellular Coverage Planning for a University Township

An operator has been licensed to provide cellular service to a 12 km² university township containing academic blocks, hostels, residential colonies and a main arterial road. The operator holds a fixed allocation equivalent to 336 duplex voice channels (after control-channel overhead) and must plan a hexagonal-cell network. Subscriber demand will grow as the township expands.

You are the radio-planning team. You must investigate how spectrum can be shared across cells, how interference limits that sharing, and how the design must change as the subscriber base grows.

Planning data and constraints

- Subscribers: 1,000 → 10,000 → 20,000 → 100,000+ (planning horizon). ● Average busy-hour traffic per subscriber: 0.025 Erlang. Target grade of service: ≤ 2% call blocking.
- Minimum acceptable SIR: choose and justify a threshold between 12 dB (digital-style) and 18 dB (analogue-style).
- Path-loss exponent n: 2.5 (open campus) to 4 (dense built-up area); justify the values you adopt.

- Spectrum cannot be increased; only the network layout and radio parameters may change.
- Any planning effort (surveys, traffic estimation, number of sites) must be included in your cost analysis.

Design Challenge (Day 4):

A convocation / sports event will bring 25,000 attendees (0.05 Erlang each) into a 1 km^2 stadium zone for four hours. Re-evaluate your original design and decide whether to retain, modify or redesign it.

TASK 1 – Design Alternative Frequency-Planning Strategies 5 Marks Guiding Enquiry

How can a fixed block of spectrum be shared across many cells so that every subscriber is served without intolerable co-channel interference — and in how many fundamentally different ways can this be done?

Investigate and interpret:

- Propose at least **three meaningfully different strategies** (for example fixed cluster reuse, sectorised reuse, dynamic/borrowed channel assignment, hierarchical macro–micro layering). The same strategy with different cluster sizes (N = 3 vs N = 7) does not count as two strategies.
- At least one strategy must be expressed as an **iterative procedure** (e.g., searching valid cluster sizes $N = i^2 + ij + j^2$ and assigning channel sets across the grid) and at least one as a **recursive/hierarchical procedure** (e.g., repeated cell splitting until demand is met).
- analyse the basic idea, the assumptions, and the way the **access channels** are shared (FDMA/TDMA/CDMA/OFDMA concept and control vs traffic channels).
- Provide pseudocode or a flowchart for each, name the design principle used (spatial reuse, trunking, spatial filtering), and justify why each deserves investigation.
- State the preprocessing required (traffic map, propagation survey, hex-grid layout) and who bears that cost.

Evidence Expected
Three distinct strategies with pseudocode/flowcharts, identified design principle, preprocessing requirements and justification for each.

TASK 2 – Identify the Governing Design Parameter 2 Marks Guiding Enquiry

For each strategy, which single quantity most strongly controls capacity and interference, and why should the analysis be built around it?

Investigate and interpret:

- Identify the dominant parameter(s) for each strategy (e.g., cluster size N, reuse ratio Q = D/R, number of co-channel interferers, cell radius R, channels per cell).
- analyse where it appears in the strategy and why it is appropriate for efficiency analysis.
- interpret how this quantity must change as subscribers grow from 1,000 to 100,000+.

Evidence Expected

Correct identification and justification of the governing parameter for each strategy, linked to its behaviour.

TASK 3 – Derive the Capacity and Interference Functions Mathematically 6 Marks

Guiding Enquiry

Starting from hexagonal geometry, how do coverage, capacity and interference depend on the design choices you made?

Investigate and interpret:

- Derive the cluster-size relation, reuse distance $D = R\sqrt{(3N)}$ and reuse ratio Q, hexagonal cell area, and the number of cells needed to cover 12 km². ● Derive channels per cell (S/N), traffic per cell at 2% blocking (Erlang-B reasoning), and total system capacity as a function of N and R.
- Derive the co-channel SIR expression for each strategy (first-tier interferers; effect of sectoring on the number of interferers).
- Combine **planning cost** (survey, number of sites) with **operating capacity** to obtain an overall cost–capacity relation for each strategy.
- Show every step; stating a final formula without derivation will not earn full credit.

Evidence Expected

Clear derivations of capacity, number of cells and SIR for each strategy, including planning cost and overall cost–capacity relation.

TASK 4 – Analyse Best, Average and Worst-Case Behaviour 4 Marks Guiding

Enquiry

Where in the cell, and at what time of day, does each strategy perform best, typically, and worst — and why?

Investigate and interpret:

- Identify the position of the mobile (near the base station, mid-cell, cell corner) that yields best, average and worst SIR, and the geometry of interferers in each case.
- Identify the traffic condition (off-peak, average, busy-hour, event surge) that yields each blocking behaviour.
- analyse how shadowing and a changing path-loss exponent shift each case. ● Support each case with calculations, not just a table of values.

Evidence Expected

Justified best/average/worst analysis for SIR and blocking, supported by mathematical reasoning.

TASK 5 – Scaling and Sensitivity Analysis 3 Marks Guiding Enquiry

How do capacity, interference and the number of cells scale as demand and the environment change, and at what point does each strategy stop being practical?

Investigate and interpret:

- Express proportionalities (e.g., capacity $\propto 1/N$, SIR $\propto N^{(n/2)}$, cells $\propto 1/R^2$) and compare growth of the strategies.
- Interpret what happens when subscribers grow from 1,000 to 100,000+ and when n varies from 2.5 to 4.
- Identify a break-even scale beyond which a different strategy becomes preferable.

Evidence Expected

Correct scaling relations and an interpretation tied to the township application.

TASK 6 – Formulate the Recurrence for Repeated Cell Splitting 3 Marks Guiding

Enquiry

If overloaded cells are split repeatedly, how can the number of cells, capacity, transmit power and handoff load after k rounds be expressed in terms of round $k - 1$?

Investigate and interpret:

- State the original problem size (cells and demand) and the subproblem produced by each split (radius $R \rightarrow R/2$).
- Identify the number of new cells per split, the work/cost per round, and the base case (the initial layout).
- Formulate recurrences for number of cells, offered capacity and required transmit power ($P' = P/2^n$), derived from your own splitting procedure. ● Allow for selective splitting (only a fraction of cells is overloaded) as well as uniform splitting.

Evidence Expected

Correct recurrence(s) with clearly identified components and base case.

TASK 7 – Solve the Recurrence and Interpret the Result 4 Marks Guiding Enquiry

How many splitting rounds are needed to carry the forecast demand, and what do the closed-form results say about cost and mobility?

Investigate and interpret:

- Solve using iteration/substitution (or another suitable method) and justify the method.
- Derive the minimum number of rounds $k = \lceil \log_4 (D/C_0) \rceil$ for uniform splitting and its selective counterpart.
- Interpret the result: growth in sites, reduction in transmit power, and the increase in handoff rate as radius shrinks.

Evidence Expected

Complete solution with method justification and an interpretation of the closed form.

TASK 8 – Implement and Test the Planning Strategies 4 Marks Guiding Enquiry

Can a simulator built from your own strategies reproduce the behaviour predicted by your mathematics?

Investigate and interpret:

- Implement all strategies (e.g., Python/NumPy): hex-grid generator, channel assignment, Monte-Carlo SIR evaluator and Erlang-B calculator. ● Use scaled datasets: 1,000 → 10,000 → 20,000 → 100,000 subscribers with the same random seeds across strategies.
- Include test cases: N = 3, 4, 7, 12; n = 2.5, 3, 4; sectored and unsectored layouts.
- Sanity-check with a known case (N = 7, n = 4 under the six-equidistant-interferer

approximation gives about 18.7 dB).

Evidence Expected

Working code, reproducible procedure and correctness checks aligned to the analysed strategies.

TASK 9 – Record and Present Experimental Performance 3 Marks Guiding Enquiry

Which measurements best expose the differences between your strategies, and how should they be presented?

Investigate and interpret:

- Record SIR distribution (mean, 5th percentile), outage probability, blocking probability, channels per cell, sites required and run-time.
- Present tables and plots such as SIR CDFs, blocking vs offered load and capacity vs N.
- Ensure the presentation allows direct comparison between strategies.

Evidence Expected

Systematically collected data presented in comparable tables and graphs.

TASK 10 – Compare Theory with Simulation 4 Marks Guiding Enquiry

Where do the simulated results agree with your analytical predictions, where do they differ, and what analyses the gap?

Investigate and interpret:

- Compare analytical and simulated SIR and blocking for each strategy. ● Identify sources of deviation: equidistant-interferer approximation, shadowing, finite tiers, mobile-position distribution, Poisson-arrival and infinite-source assumptions in Erlang-B, integer rounding of N and channels.
- Decide which strategy scales best and defend this using both results.

Evidence Expected

A reasoned comparison of theory and simulation with analyseed discrepancies.

TASK 11 – Recommend and Defend the Final Design 4 Marks Guiding Enquiry

Which design would you actually deploy in the township, and what evidence makes it better than the alternatives?

Investigate and interpret:

- State the recommended strategy, parameters (N, R, sectoring) and expected capacity/SIR/blocking.
- analyse why it is superior to the alternatives using your analysis and simulation.
- interpret planning cost, trade-offs (trunking efficiency vs interference, handoff load vs capacity) and limitations.
- Judge whether the design remains practical as the township grows past 100,000 subscribers.

Evidence Expected

A technically justified recommendation supported by analysis and simulation; choosing the highest-capacity option alone is not sufficient.

DAY-4 DESIGN CHALLENGE

TASK 12 – Re-evaluate and Adapt the Design 8 Marks Changed Requirement

A convocation / sports event will bring 25,000 attendees, each generating 0.05 Erlang, into a 1 km² stadium zone for four hours. Outside the event, demand stays as originally forecast. Event days occur roughly twice a month.

Re-evaluate your original design under the changed conditions.

You may:

RETAIN → MODIFY → or REDESIGN

your original solution.

A. Impact Analysis – 2 Marks

Guiding Enquiry

What exactly has changed, and which of your original assumptions no longer hold?

- Estimate the new offered traffic density and compare it with the design density. ● Identify affected assumptions (uniform traffic, steady demand, permanent capacity).
- analyse why the short duration and predictability of the surge matter for the choice of solution.

Evidence Expected

A clear analysis of how the surge affects the original design.

B. Evaluation of the Original Design – 2 Marks

Guiding Enquiry

How would your recommended design behave on event day?

- Estimate blocking probability and interference problems in the stadium zone using your own model.
- Assess whether the design can be reused unchanged and what fails first. ● Identify whether an alternative strategy handles the surge more economically.

Evidence Expected

A technically supported evaluation of the original design under the surge. **C.**

Adaptation or Redesign – 2 Marks

Guiding Enquiry

Should the township network be altered permanently, temporarily or not at all?

- Choose retain, modify or redesign (e.g., micro-cells/cell splitting in the stadium zone, 60° sectoring, temporary cells-on-wheels, dynamic channel borrowing). ● Describe the revised strategy and what changes in reuse pattern, power and handoffs.
- Determine the effect on capacity, SIR and planning cost.

Evidence Expected

A design response with clear radio-planning reasoning and quantified effect. **D.**

Evidence-Based Defence – 2 Marks

Guiding Enquiry

Why is your final decision the right one, given cost, frequency of events and the evidence you generated?

- Use analytical results, simulation results, surge size and duration, planning cost and trade-offs.
- Show that the decision follows logically from your evidence.

Evidence Expected

A well-supported final decision integrating theory, simulation and the changed constraints.

ASSIGNMENT-1 – ANSWERS

Planning assumptions used throughout the answers

The township area is 12 km² and the available spectrum is fixed at 336 duplex traffic channels. Busy-hour offered traffic per subscriber is 0.025 Erlang and the target blocking probability is 2%. I select a minimum SIR target of 18 dB because it gives a conservative interference margin. For open campus areas I use path-loss exponent n = 2.5–3; for dense built-up areas I use n = 4. Unless otherwise stated, calculations use N = 7, six first-tier co-channel interferers, and the standard hexagonal-cell approximation. Numerical site and cost values are planning estimates, not vendor quotations.

TASK 1 – Design Alternative Frequency-Planning Strategies (5 Marks)

Strategy 1: Fixed cluster frequency reuse

The 336 traffic channels are divided into N channel groups and the groups are repeated across a hexagonal grid. With N = 7, each cell receives S/N = 336/7 = 48 channels. A cluster contains N cells and the same channel group is reused only after the reuse distance D. This is spatial reuse and is appropriate when traffic is relatively stable and interference must be predictable.
Hexagonal cluster sizes must satisfy $N = i^2 + ij + j^2$, where i and j are non-negative integers. The procedure below searches valid cluster sizes and chooses the smallest one meeting the SIR target.
Pseudocode:
Input: total_channels S=336, SIR_target=18 dB, path_loss n, cell radius R
Generate candidate N = i*i + i*j + j*j for integer i,j
For each N: channels_per_cell = floor(S/N)
Compute D/R = sqrt(3*N) and SIR = (D/R)^n / 6
Reject candidates whose SIR is below the target
Choose the smallest remaining N with adequate capacity; assign each channel group to N cells
Repeat the channel groups over the hexagonal grid
Preprocessing: prepare a subscriber/traffic heat map, measure propagation and shadowing, select a hexagonal grid, and verify candidate sites. The operator should pay for radio surveys, traffic modelling and site planning because these are part of network deployment.

Strategy 2: Sectorised cells

Each base station uses directional antennas instead of one omnidirectional antenna. A 120° layout has three sectors and a 60° layout has six sectors. Directional antennas reduce the number of dominant co-channel interferers seen by a sector. The design principle is spatial filtering. The channel pool is still fixed; channel groups are assigned to sectors so adjacent sectors and nearby co-channel sectors do not use interfering groups.
Pseudocode:
Input: hex grid, S=336, candidate N, sector count = 3 or 6
Divide each cell into sectors and assign channel groups by sector orientation
For sampled mobile positions, calculate desired signal and directional interference
Estimate the 5th-percentile SIR and outage probability
Increase N or change sector orientation if the SIR target is not met
Keep the sector plan with acceptable SIR and required capacity
Preprocessing: in addition to the traffic map and propagation survey, obtain antenna patterns, azimuths and tilt information. Sectorisation adds antenna and installation costs but can improve SIR without buying extra spectrum.

Strategy 3: Dynamic channel assignment / channel borrowing with hierarchical cell splitting

Channels are selected from a common eligible pool according to current interference and traffic rather than being permanently fixed to every cell. A lightly loaded cell may lend an unused channel to an overloaded neighbour if the reuse and interference constraints remain satisfied. For long-term growth, overloaded macro-cells are recursively split into smaller micro-cells. The principles are trunking efficiency, dynamic allocation and hierarchical reuse.
Pseudocode:
For each call request, list channels that are free and satisfy the interference constraint
Allocate a locally preferred channel; if none exists, request a safe channel from a neighbour
If no safe channel is available, block the call and record the event
At each planning interval, identify cells exceeding the traffic threshold
Split only overloaded cells, reduce radius, add sites and recompute channel assignment
Stop splitting when blocking is at most 2% or a cost/site limit is reached
Preprocessing: time-stamped busy-hour traffic, interference measurements, neighbour-cell data, candidate micro-site survey and a handoff map are needed. The operator pays for planning and the extra control software; borrowing can improve use of existing channels but requires coordination.

Access-channel interpretation

FDMA separates users by frequency; TDMA separates users by time slots; CDMA separates users by spreading codes; OFDMA assigns groups of orthogonal subcarriers and is common in modern cellular systems. The 336-channel figure is treated as the available duplex traffic-channel equivalent after control-channel overhead. Broadcast, synchronisation, paging

and access/control resources must be reserved separately in a real system; they must not be counted again as traffic channels.

TASK 2 – Identify the Governing Design Parameter (2 Marks)

Fixed reuse: the governing parameter is cluster size N. Increasing N increases D/R = √(3N) and improves SIR, but reduces channels per cell from 336/N. Sectorisation: the main parameters are sector angle and the number of dominant co-channel interferers i_0; narrower sectors generally reduce interference but increase antenna and planning complexity. Dynamic borrowing: the governing parameters are the number of free eligible channels and the blocking/interference threshold. Cell splitting: radius R and cells per area govern spatial capacity and site count.
As demand grows from 1,000 to 100,000+ subscribers, the design must add spatial reuse by splitting overloaded cells, improve sectorisation where interference is dominant, and use dynamic assignment to reduce avoidable blocking. N cannot simply be made very small if the 18 dB SIR target is violated.

TASK 3 – Derive the Capacity and Interference Functions Mathematically (6 Marks)

1. Hexagonal geometry and reuse

For a regular hexagonal cell of centre-to-vertex radius R,
the area is $A_{cell} = (3\sqrt{3}/2)R^2 \approx 2.598R^2$ km². The approximate number of cells covering area A_T is $C = \text{ceil}(A_T/A_{cell})$. For R = 0.5 km,
$A_{cell} = 2.598(0.5^2) = 0.6495$ km² and $C = \text{ceil}(12/0.6495) = 19$ cells.
A valid cluster size is $N = i^2 + ij + j^2$. The co-channel reuse distance is $D = R\sqrt{(3N)}$, so reuse ratio $Q = D/R = \sqrt{(3N)}$.
For N = 7, $Q = \sqrt{21} = 4.583$ and $D = 0.5(4.583) = 2.291$ km.

2. Channels and Erlang-B capacity

Channels per cell m = S/N = 336/7 = 48.
Erlang-B blocking is B(m,A) = [A^m/m!] / Σ(k=0 to m)[A^k/k!],
where A is offered traffic in Erlangs. It can be calculated stably using $B_0 = 1$ and $B_j = (A\ B_{(j-1)})/(j + A\ B_{(j-1)})$. For m = 48, B(48,38) ≈ 1.80%, while B(48,39) is above 2%; therefore approximately 38 Erlangs per cell is the planning limit at 2% blocking.
Total offered traffic is $A_{total} = U \times 0.025$ Erlang.
Thus 1,000 users offer 25 Erlangs; 10,000 offer 250; 20,000 offer 500; and 100,000 offer 2,500 Erlangs. A rough capacity bound is $A_{capacity} \approx C \times 38$ Erlangs, subject to traffic distribution and interference constraints.

3. Co-channel SIR

For equal-power base stations and path-loss exponent n, SIR = $S/(\Sigma I_j)$. Under the six-equidistant first-tier interferer approximation, each interferer is at distance D and the

desired mobile is approximated at distance R, giving SIR ≈ (D/R)^n/6 = (3N)^(n/2)/6. For N = 7 and n = 4: SIR = $21^2/6$ = 73.5, or $10\log_{10}(73.5)$ = 18.66 dB. This agrees with the assignment's sanity-check value of about 18.7 dB. For n = 2.5, the same approximation gives about 10.7 dB, illustrating that open environments can have worse co-channel SIR under this simplified model.

For sectoring, replace six equal interferers with the smaller set of interferers that fall within the serving sector's antenna pattern. A simplified expression is SIR ≈ (D/R)^n/i_eff, where i_eff is the effective number of interferers after antenna attenuation. Real analysis should weight each interferer by its antenna gain and actual distance.

4. Cost–capacity relationship

Let c_site be the installed cost per site, C the number of sites, c_survey the survey cost, and c_op the operating cost per site per planning period. Then C_plan = c_survey + C(c_site + c_op). A useful cost-efficiency measure is cost per carried Erlang: J = C_plan / A_carried, where A_carried = A_offered(1 − B) for a uniform model. Fixed reuse has lower complexity but fixed channel groups; sectorisation adds antenna cost; dynamic borrowing adds coordination cost; cell splitting adds site and backhaul costs but increases spatial capacity.

Illustrative site planning: if c_site = ₹8,00,000, c_survey = ₹2,00,000 and C = 19, initial capital planning cost is ₹1.54 crore (₹2 lakh + 19 × ₹8 lakh), excluding spectrum, backhaul and recurring operations. These are assumptions for comparison, not quoted prices.

TASK 4 – Analyse Best, Average and Worst-Case Behaviour (4 Marks)

Best SIR occurs near the serving base station, where desired signal power is high, and where the mobile lies away from the strongest co-channel directions. Average SIR occurs at a representative mid-cell position with ordinary shadowing. Worst SIR occurs near a cell edge/corner, especially when several co-channel cells are geometrically close and shadowing weakens the desired signal.

Using a path-loss model P ∝ d^(−n), the desired-to-interference ratio changes with the mobile's serving distance and its distances to interferers. At the cell edge, desired power falls as R^(−n), while one or more interferers may be relatively close; therefore SIR decreases. With N = 7, n = 4 and six equidistant interferers, the reference SIR is 18.66 dB. If the effective interferer count doubles from 6 to 12 while distances stay fixed, SIR halves: $10\log_{10}(73.5/2)$ = 15.65 dB.

Off-peak traffic normally has low blocking because offered traffic A is small. Average-hour blocking is moderate. Busy-hour or event-surge blocking is highest because A approaches or exceeds the Erlang-B limit. With 48 channels, A = 38 Erlangs gives approximately 1.80% blocking; at A = 40 Erlangs, blocking is higher than 2%. Shadowing can create local dead spots and raise outage probability. A larger path-loss exponent reduces distant interference more strongly, often improving reuse SIR, but dense buildings may also create obstruction and shadowing, so measurements are essential.

TASK 5 – Scaling and Sensitivity Analysis (3 Marks)

For fixed N and channel pool S, channels per cell are approximately S/N, so per-cell capacity decreases approximately as 1/N. Reuse distance is D/R = √(3N), and under the six-interferer approximation SIR ∝ N^(n/2). Cell count scales as C ∝ 1/R². Halving R creates about four times as many cells over the same area.

Subscriber traffic grows linearly: 1,000 → 100,000 users increases offered traffic from 25 to 2,500 Erlangs, a factor of 100. At 38 Erlangs per cell, the idealised cell counts needed are ceil(25/38)=1, ceil(250/38)=7, ceil(500/38)=14 and ceil(2500/38)=66; practical coverage and geographic distribution can require more sites. At n = 2.5, the N = 7 simplified SIR is only about 10.7 dB, while at n = 4 it is 18.7 dB. Thus sectoring and measured propagation are important in open areas. Fixed reuse alone becomes costly or inflexible at high demand; dynamic borrowing helps short-term imbalances, while selective micro-cell splitting is preferable when overload is persistent.

TASK 6 – Formulate the Recurrence for Repeated Cell Splitting (3 Marks)

Let C_0 be the original number of cells, R_0 the original radius, and D_0 the offered traffic. One uniform split round reduces radius from R to R/2. Since cell area is proportional to R², each parent cell is replaced by approximately four smaller cells. Therefore C_k = 4C_(k−1), C_0 given, and hence C_k = $C_0$4^k. The radius is R_k = R_0/2^k.

If the same channel reuse pattern is maintained, the idealised area capacity scales with cell count: K_k = 4K_(k−1) = $K_0$4^k. This assumes each new cell can use the same per-cell channel allocation without violating interference constraints. With the assignment's power model, P_k = P_0/2^(nk), because P′ = P/2^n at each split. Smaller cells create more boundaries, so handoff attempts and mobility-management load increase.

For selective splitting, let f_k be the fraction of cells split in round k. Then C_k = C_(k−1)[1 + 3f_k], because each selected parent is replaced by four children, adding three cells. For a constant fraction f, C_k = C_0(1+3f)^k. The base case is k=0: C=C_0, R=R_0, P=P_0 and capacity K=K_0. Planning cost per round includes new sites, backhaul, power, permits and optimisation.

TASK 7 – Solve the Recurrence and Interpret the Result (4 Marks)

For uniform splitting, C_k = 4C_(k−1). Substitution gives C_1=4C_0, C_2=4(4C_0)=4²C_0, and by induction C_k=$C_0$4^k. If original capacity is K_0 and demand is D, require $K_0$4^k ≥ D. Dividing by K_0 gives 4^k ≥ D/K_0; taking log base 4 gives k ≥ $\log_4$(D/K_0). The minimum integer number of rounds is k = max(0, ceil($\log_4$(D/K_0))).

Example: if the current layout carries 700 Erlangs and forecast demand is 2,500 Erlangs, k = ceil($\log_4$(2500/700)) = 1 round. The idealised capacity becomes 2,800 Erlangs. Sites become four times the original count and radius halves. This is a capacity approximation; channel availability, interference, and nonuniform traffic must be checked.

For selective splitting with constant split fraction f, capacity multiplier per round is approximately (1+3f), so k = max(0, ceil(log(D/K_0)/log(1+3f))). If f=0, no capacity is added.

More splitting increases site/backhaul cost and handoff rate; selective splitting avoids spending money in low-demand areas.

TASK 8 – Implement and Test the Planning Strategies (4 Marks)

The following Python program is a compact reproducible baseline. It calculates valid cluster sizes, channels per cell, hexagonal cell count, Erlang-B blocking, and the six-interferer analytical SIR. It uses a fixed random seed for reproducibility. It is an analytical planning simulator; a full Monte-Carlo spatial simulator would additionally generate mobile coordinates, shadowing samples, antenna patterns and individual interferer distances.

```
import math
import random

random.seed(7)

def erlang_b(channels, traffic):
    b = 1.0
    for m in range(1, channels + 1):
        b = traffic * b / (m + traffic * b)
    return b

def valid_clusters(limit=20):
    values = set()
    for i in range(limit + 1):
        for j in range(limit + 1):
            n = i*i + i*j + j*j
            if n > 0:
                values.add(n)
    return sorted(values)

def hex_area(radius):
    return (3 * math.sqrt(3) / 2) * radius * radius

def sir_db(n_cluster, path_loss=4.0, interferers=6):
    sir = (3 * n_cluster) ** (path_loss / 2) / interferers
    return 10 * math.log10(sir)

S = 336
radius = 0.5
area = 12
users = [1000, 10000, 20000, 100000]

print("N, channels/cell, SIR dB, cells, traffic, blocking")
```

```
for N in [3, 4, 7, 12]:
    channels = S // N
    cells = math.ceil(area / hex_area(radius))
    for U in users:
        traffic = U * 0.025 / cells
        blocking = erlang_b(channels, traffic)
        print(N, channels, round(sir_db(N), 2), cells,
              round(traffic, 2), round(blocking * 100, 2))

print("Valid cluster sizes:", valid_clusters())
```

Test cases required by the assignment are N = 3, 4, 7, 12; n = 2.5, 3, 4; sectorised and unsectorised layouts; and subscriber levels 1,000, 10,000, 20,000 and 100,000. The known check N=7, n=4 should produce about 18.66 dB. N=3 gives a lower SIR under the same approximation and should be flagged if it does not meet the selected 18 dB target.

TASK 9 – Record and Present Experimental Performance (3 Marks)

Record, for every strategy and each traffic level: mean SIR, 5th-percentile SIR, outage probability (fraction below 18 dB), Erlang-B blocking probability, channels per cell, required sites and execution time. Keep the same random seed and mobile positions for fair comparisons.

Suggested results table: Strategy | N | sector count | path-loss n | mean SIR (dB) | 5th-percentile SIR (dB) | outage (%) | blocking (%) | sites | runtime (s). Plot (1) SIR cumulative distribution functions, (2) blocking probability versus offered traffic, and (3) ideal capacity versus N. Label units, include assumptions and use identical axes when comparing strategies.

For the analytical reference with N=7 and n=4, SIR is 18.66 dB. With 48 channels, Erlang-B blocking is approximately 1.80% at 38 Erlangs per cell. These are reference values for validating the implementation, not claimed outputs from a full spatial Monte-Carlo experiment.

TASK 10 – Compare Theory with Simulation (4 Marks)

Theory and simulation should agree closely when the simulation uses six equal-distance interferers, equal transmit powers, no shadowing, and a mobile at the assumed reference position. The N=7, n=4 reference should be about 18.7 dB. Erlang-B should also match when the simulator uses the same number of channels and offered traffic.

Differences arise because real mobiles are not all at the same distance; the number and distances of interferers vary; shadowing and antenna patterns alter received powers; only a finite number of cell tiers may be simulated; and random sampling produces estimation error. Erlang-B assumes a loss system with Poisson call arrivals, independent call holding times and an infinite-source approximation. Rounding N and channels to integers also changes the result.

The likely scalable choice is a hybrid: N=7 fixed reuse as a robust baseline, sectoring to reduce interference, dynamic borrowing for temporary local imbalance, and selective splitting for sustained high demand. A pure N=3 layout has more channels per cell but fails the conservative SIR target in the simplified model; pure fixed assignment can waste channels when traffic is uneven.

TASK 11 – Recommend and Defend the Final Design (4 Marks)

Recommended township design: use N=7 as the baseline reuse cluster, target SIR ≥18 dB, use three-sector (120°) antennas at high-interference/high-demand sites, and apply selective micro-cell splitting in dense hostels, academic blocks and road corridors. Use dynamic channel borrowing only when a borrowed channel passes the interference check. Begin with R≈0.5 km for the initial layout, then densify as traffic maps show overload.

For R=0.5 km, 19 cells approximately cover 12 km². Each N=7 cell has 48 channels and a planning limit near 38 Erlangs at 2% blocking, giving an idealised total of about 722 Erlangs. This is enough for the 1,000-user load (25 Erlangs) and, in a uniform ideal model, 10,000 users (250 Erlangs), but not 20,000 users (500 Erlangs) if coverage, busy-hour imbalance and reserve margins are considered; 100,000 users require roughly ceil(2500/38)=66 traffic-capable cells as an ideal lower bound. At R=0.25 km, about 74 cells cover 12 km² and the idealised capacity is about 2,812 Erlangs.

The design balances interference, capacity, handoffs and cost. Increasing N improves SIR but lowers channels per cell; sectoring improves spatial isolation but adds antennas; splitting adds sites and backhaul and increases handoffs. Beyond 100,000 users, a macro-only design is not practical without dense small cells, careful backhaul planning and continued traffic measurements. Final values must be validated by a propagation survey and detailed simulation.

TASK 12 – Re-evaluate and Adapt the Design: Day-4 Challenge (8 Marks)

A. Impact Analysis (2 Marks)

Event traffic is 25,000 attendees × 0.05 Erlang/person = 1,250 Erlangs concentrated in 1 km² for four hours. The event traffic density is 1,250 Erlangs/km². For comparison, the 100,000-user township average is 2,500/12 = 208.3 Erlangs/km², so the stadium event density is about six times that average. Compared with the 20,000-user township average (500/12 = 41.7 Erlangs/km²), it is about 30 times higher.

The assumptions of uniform traffic, steady demand and permanent capacity no longer hold. The event is localised, short, large and predictable, so a temporary solution can be more economical than permanently densifying the entire township.

B. Evaluation of the Original Design (2 Marks)

At R=0.25 km, cell area is 0.1624 km², so approximately 1/0.1624 = 6.16 cell-equivalents lie in the 1 km² stadium zone. At about 38 Erlangs per 48-channel cell, this gives only about 234

Erlangs of idealised traffic capacity, far below 1,250 Erlangs. The original design therefore cannot carry the event traffic with a 2% blocking target. Blocking and congestion would rise first; interference and access-channel congestion may also increase as users concentrate.
A simple lower-bound cell count is ceil(1250/38)=33 cells with 48 channels per cell and 2% blocking target. This excludes reserve capacity, traffic imbalance, control-channel limits and interference constraints, so an operational design should provide margin beyond 33 cell-equivalents.

C. Adaptation or Redesign (2 Marks)

Decision: MODIFY temporarily rather than permanently redesign the whole township. Deploy temporary cells-on-wheels or compact temporary micro-cells around the stadium, use sectorised antennas and interference-aware dynamic channel borrowing, and tune transmit power to avoid excessive overlap. Maintain N=7 as the starting reuse plan; adjust local channel groups and sector orientation after an event-zone interference check.
Plan for at least 33 idealised 48-channel cell-equivalents to meet 1,250 Erlangs at approximately 2% blocking; add a practical reserve based on measured load and outages. The temporary sites increase local capacity and improve SIR through shorter serving distances and sectoring, but they also increase handoffs and require backhaul, power, permits and on-site coordination. A planning budget should include event surveys, temporary equipment rental, installation, transport and four-hour operation. A permanent build-out would incur recurring cost despite events occurring only about twice a month.

D. Evidence-Based Defence (2 Marks)

Temporary modification is the best trade-off because the surge is predictable, concentrated in one square kilometre and lasts only four hours, about twice monthly. A permanent network sized for 1,250 Erlangs/km² would overbuild most of the township during normal days. Temporary cells-on-wheels and sectorised micro-cells put capacity where and when it is needed, while dynamic borrowing can use spare channels from neighbouring cells if interference permits.
The decision is supported by the calculation: 1,250 Erlangs are offered in the stadium zone, while approximately six existing cell-equivalents provide only about 234 Erlangs at the 2% target. At least 33 idealised cell-equivalents are needed before adding reserve margin. The revised design should be accepted only after the simulator verifies SIR/outage and blocking for the event traffic map. The main limitations are the simplified uniform traffic assumption, approximate Erlang-B model, uncertain shadowing, and the need to verify real antenna patterns and temporary-site backhaul.

Overall conclusion

A fixed spectrum can support a growing township through spatial reuse, sectorisation, dynamic channel assignment and selective cell splitting. N=7 provides an approximately 18.7 dB reference SIR when n=4 under the six-interferer approximation, while 48 channels per cell support about 38 Erlangs at 2% blocking. Traffic growth requires additional cells and careful

reuse planning; the stadium surge requires temporary local capacity rather than permanent township-wide overbuilding. All final site counts and cost figures should be refined using field surveys, measured traffic and a full spatial simulation.

MARK DISTRIBUTION

Task	**Marks**
Task 1 – Design Alternative Frequency Planning Strategies	5
Task 2 – Identify the Governing Design Parameter	2
Task 3 – Derive the Capacity and Interference Functions Mathematically	6
Task 4 – Analyse Best, Average and Worst Case Behaviour	4
Task 5 – Scaling and Sensitivity Analysis	3
Task 6 – Formulate the Recurrence for Repeated Cell Splitting	3
Task 7 – Solve the Recurrence and Interpret the Result	4
Task 8 – Implement and Test the Planning Strategies	4

Task 9 – Record and Present Experimental Performance	3
Task 10 – Compare Theory with Simulation	4
Task 11 – Recommend and Defend the Final Design	4
Subtotal	**42**
Task 12 – Day-4 Design Challenge	8
TOTAL	**50**